

Experiment 20: Planets Database Using Prolog

Aim

To develop a **Planets Database using Prolog** and retrieve information about planets using facts, rules, and queries.

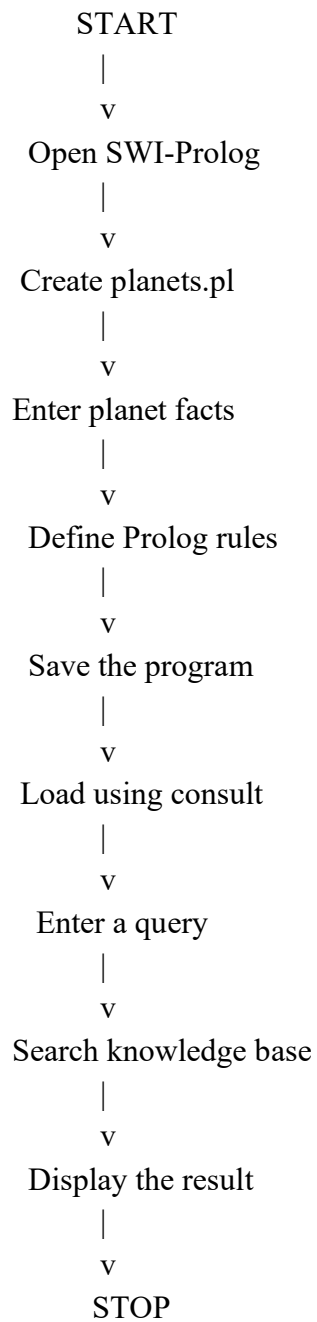
Objective

1. To represent planetary information using Prolog facts.
2. To classify planets into terrestrial, gas giant, and ice giant planets.
3. To understand the use of variables and rules in Prolog.
4. To retrieve required information using Prolog queries.

Algorithm

1. Start the SWI-Prolog environment.
2. Create a Prolog file named planets.pl.
3. Define facts containing the planet name, type, and distance from the Sun.
4. Define rules to identify:
 - a. Terrestrial planets
 - b. Gas giant planets
 - c. Ice giant planets
5. Save the program as planets.pl.
6. Load the file into SWI-Prolog using consult.
7. Execute queries to retrieve planet information.
8. Display the results.
9. Stop.

Flowchart



Prolog Program

% Planets Database

```
planet(mercury, terrestrial, 57.9).  
planet(venus, terrestrial, 108.2).  
planet(earth, terrestrial, 149.6).  
planet(mars, terrestrial, 227.9).  
planet(jupiter, gas_giant, 778.5).  
planet(saturn, gas_giant, 1434).  
planet(uranus, ice_giant, 2871).  
planet(neptune, ice_giant, 4495).
```

% Rule to find terrestrial planets
terrestrial_planet(X) :-
 planet(X, terrestrial, _).

% Rule to find gas giant planets
gas_giant_planet(X) :-
 planet(X, gas_giant, _).

% Rule to find ice giant planets
ice_giant_planet(X) :-
 planet(X, ice_giant, _).

Sample Output

Query 1

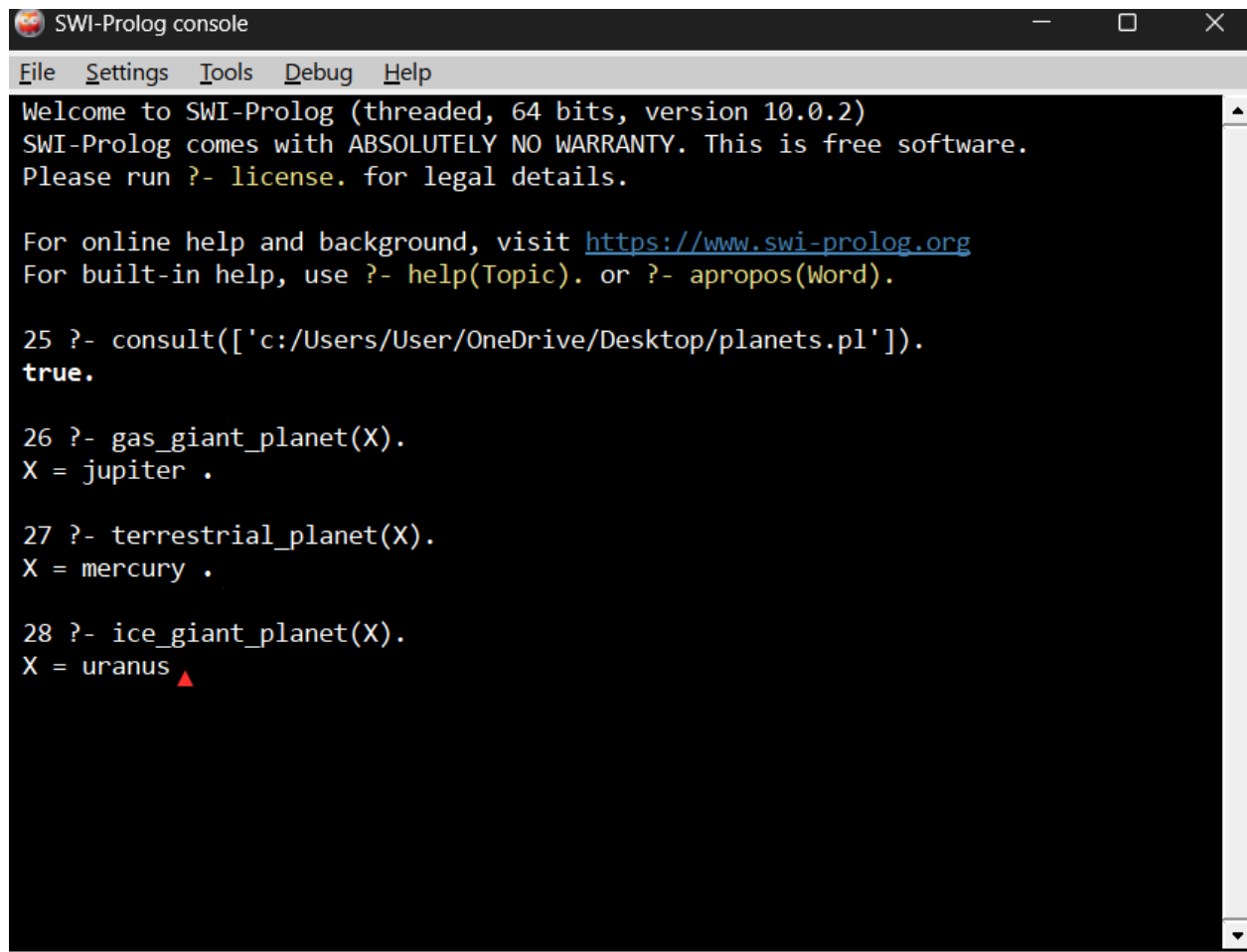
?- terrestrial_planet(X).

Query 2

?- gas_giant_planet(X).

Query 3

?- ice_giant_planet(X).



```
SWI-Prolog console
File Settings Tools Debug Help
Welcome to SWI-Prolog (threaded, 64 bits, version 10.0.2)
SWI-Prolog comes with ABSOLUTELY NO WARRANTY. This is free software.
Please run ?- license. for legal details.

For online help and background, visit https://www.swi-prolog.org
For built-in help, use ?- help(Topic). or ?- apropos(Word).

25 ?- consult(['c:/Users/User/OneDrive/Desktop/planets.pl']).
true.

26 ?- gas_giant_planet(X).
X = jupiter .

27 ?- terrestrial_planet(X).
X = mercury .

28 ?- ice_giant_planet(X).
X = uranus ▲
```

Result

The **Planets Database** was successfully implemented in **Prolog**, and the required information about planets was retrieved correctly using facts, rules, and queries.

Conclusion

Thus, the experiment successfully demonstrates how **Prolog can be used to represent and retrieve knowledge about planets**. The database can classify planets based on their type and provide information such as their distance from the Sun.